

Asynchronous Analog-Compute Edge Accelerator

Author: Nachiket Kumar

Department: Electrical and Electronics Engineering (EEE)

Institution: Bakhtiyarpur College of Engineering (BCE)

Target Application: Non-Intrusive Load Monitoring (NILM)

1. Abstract

Modern electrical fingerprinting and Non-Intrusive Load Monitoring (NILM) require high-speed analysis of nonlinear load signatures. Standard digital GPUs suffer from excessive power overhead due to rigid clock trees and digital memory bottlenecks. This portfolio documents the architectural design, Register-Transfer Level (RTL) Verilog implementation, and behavioral simulation of a novel hybrid ASIC. The architecture fuses a clockless, asynchronous Network-on-Chip (NoC) pipeline with a physics-based analog-compute execution core, performing Ohm's Law ($I = V \times G$) for instant matrix multiplications.

2. Asynchronous Routing: The Muller C-Element

To eliminate the global clock, data propagation is governed by localized 4-phase handshakes utilizing Muller C-elements. This ensures power is only drawn when electrical transient data is actively moving through the NoC.

```
module muller_c (  
    input wire a,          // Request  
    input wire b,          // Acknowledge  
    input wire rst_n,  
    output reg y  
);  
    always @(a or b or rst_n) begin  
        if (!rst_n) y <= 1'b0;  
        else if (a == 1'b1 && b == 1'b1) y <= 1'b1;  
        else if (a == 1'b0 && b == 1'b0) y <= 1'b0;  
    end  
endmodule
```

3. The Analog Compute Core

The execution engine bypasses standard digital ALUs. The behavioral model simulates a memristor crossbar array. High-frequency electrical inputs are treated as voltages (**V**), matrix weights as conductances (**G**), and the parallel FFT-style calculations are executed physically as currents (**I**). Physical noise and precision limits are modeled via bitwise truncation.

```
// Ohm's Law Matrix Math
raw_i_0 = (v_in_0 * G_00) + (v_in_1 * G_10);
raw_i_1 = (v_in_0 * G_01) + (v_in_1 * G_11);

// Analog Noise Simulation (Truncating LSB)
i_out_0 = raw_i_0[WIDTH-1:0] & 8'b11111110;
i_out_1 = raw_i_1[WIDTH-1:0] & 8'b11111110;
```

4. Behavioral Simulation & Waveforms

The integrated `signal\_gpu\_top` module was simulated using Icarus Verilog and GTKWave. A dual-channel electrical transient was injected (Hex 0A and 05). The asynchronous NoC successfully routed the packet upon the `req\_in` trigger. The analog core computed the matrix operation, returning Hex 18 (Dec 24), accurately reflecting the designed hardware precision loss.

[Insert GTKWave 'master_wave.vcd' Screenshot Here]

Figure 1: GTKWave output confirming clockless propagation and behavioral analog calculation.

5. Future Work: Synthesis & Scalability

The immediate next phase involves mapping this RTL to an FPGA using Xilinx Vivado to extract power estimations and Look-Up Table (LUT) utilization. Following synthesis verification, the

analog compute wrapper will be algorithmically scaled via Python scripting from a 2x2 matrix to a dense 16x16 crossbar array, capable of deeply analyzing 16 concurrent harmonic frequency channels for predictive maintenance and edge classification.